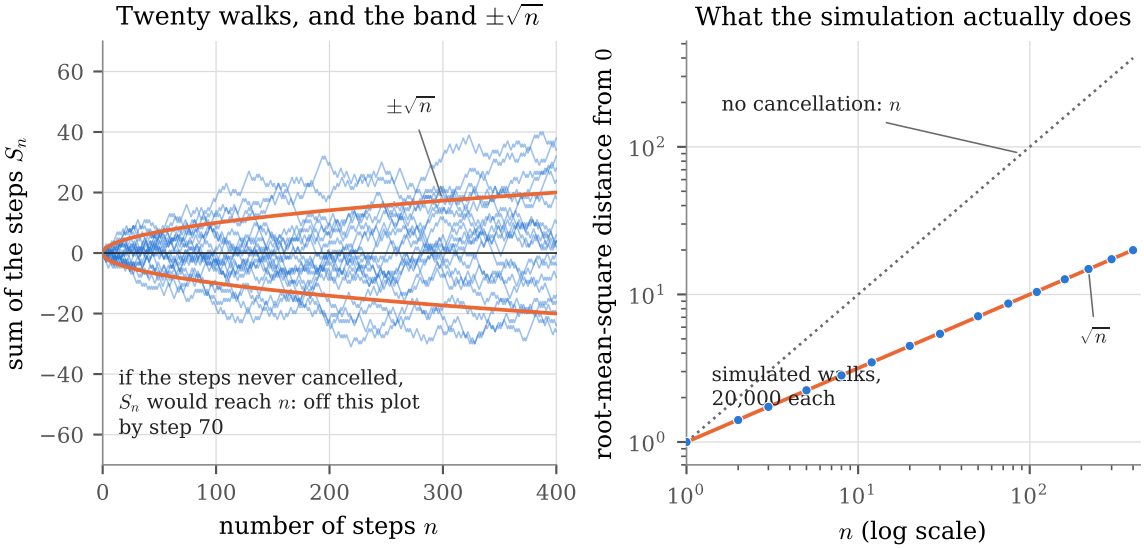
Twenty walks, and the band $\pm\sqrt{n}$
$\pm\sqrt{n}$
if the steps never cancelled,
S_n would reach n: off this plot
by step 70
sum of the steps S_n
number of steps n
60
40
20
0
−20
−40
−60
0
100
200
300
400
What the simulation actually does
no cancellation: n
$\sqrt{n}$
simulated walks,
20,000 each
root-mean-square distance from 0
n (log scale)
10^0
10^1
10^2